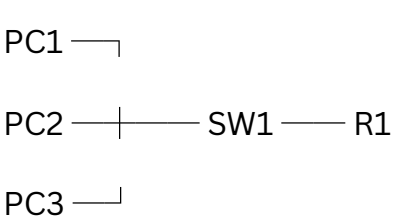
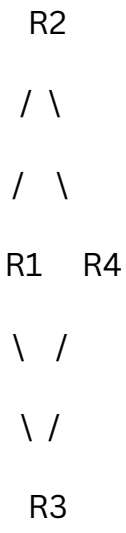
Phase 1 progress:

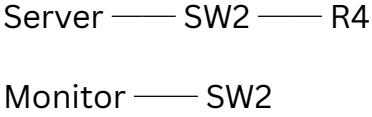
Here's the breakdown:

Phase 1 task	Status
Add 4 routers	✅
Add 2 switches	✅
Add 3 clients	✅
Add Server + Monitor-PC	✅
Physical connections / ports	✅
Router IP addressing	✅
Router interfaces up/up	✅
Client LAN configuration	✅
Server LAN configuration	✅
Basic LAN ping tests	✅
Configure OSPF	⬜
Verify OSPF neighbors	⬜
Verify routing tables	⬜
End-to-end client → server ping	⬜
Test R2-R4 failure	⬜
Verify OSPF failover R1→R3→R4	⬜
Restore and retest network	⬜

What we've established so far



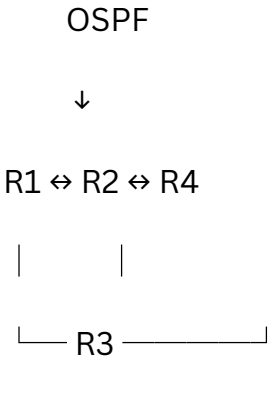




All the **physical links and IP addressing are working**. That's the foundation.

The remaining ~60% is the important networking part

Next we'll configure:



End-to-end connectivity

↓

Break R2-R4

↓

OSPF detects failure

↓

R1 → R3 → R4

↓

Connectivity restored

Next step: OSPF configuration. We'll configure **R1 first, check it, then R2, R3, and R4**, rather than pasting everything at once.

Continue with the network setup

- Check the remaining LAN connectivity
- Configure OSPF on R1

i want to create a report file of what all we have done so far also give me all the configuration and port connections and topology so that i can add them to a file

Yes. Since we're building this phase-by-phase, it's a good idea to maintain a **project report/log** from the beginning. Below is a clean section you can directly put into a Markdown file such as:

docs/phase1_progress.md

It records **only what we've actually completed so far**. OSPF configuration and failover are not included as completed yet.

DNSentinel — Phase 1 Progress Report

1. Project Information

Project Name: DNSentinel

Project Title: Adaptive Encrypted DNS Tunnel with Multi-Client Reliable Transfer, Congestion Control & OSPF Failover

Phase: Phase 1 — Cisco Network & OSPF

Current Status: Network topology and IP addressing completed; OSPF configuration pending.

2. Objective of Phase 1

The objective of Phase 1 is to establish the underlying computer network that will later support the DNSentinel system.

The network consists of:

- 4 routers
- 2 switches
- 3 client PCs
- 1 DNS/Tunnel Server
- 1 Monitor PC
- Redundant router paths
- OSPF dynamic routing
- OSPF-based link failover

The network is designed so that there are two possible paths between the client-side router R1 and server-side router R4.

Normal path

R1 → R2 → R4

Backup path

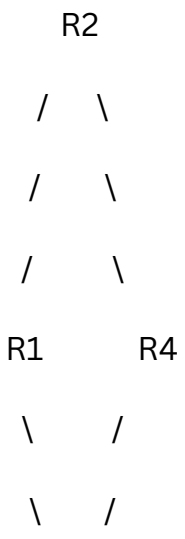
R1 → R3 → R4

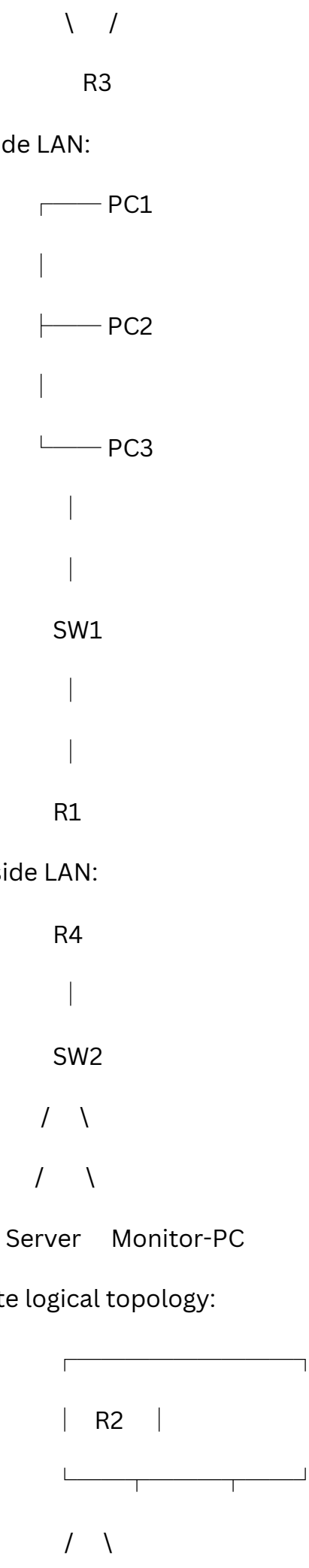
If the R2-R4 link fails, OSPF should automatically select the alternate path:

R1 → R3 → R4

3. Network Topology

The physical topology designed for the project is:

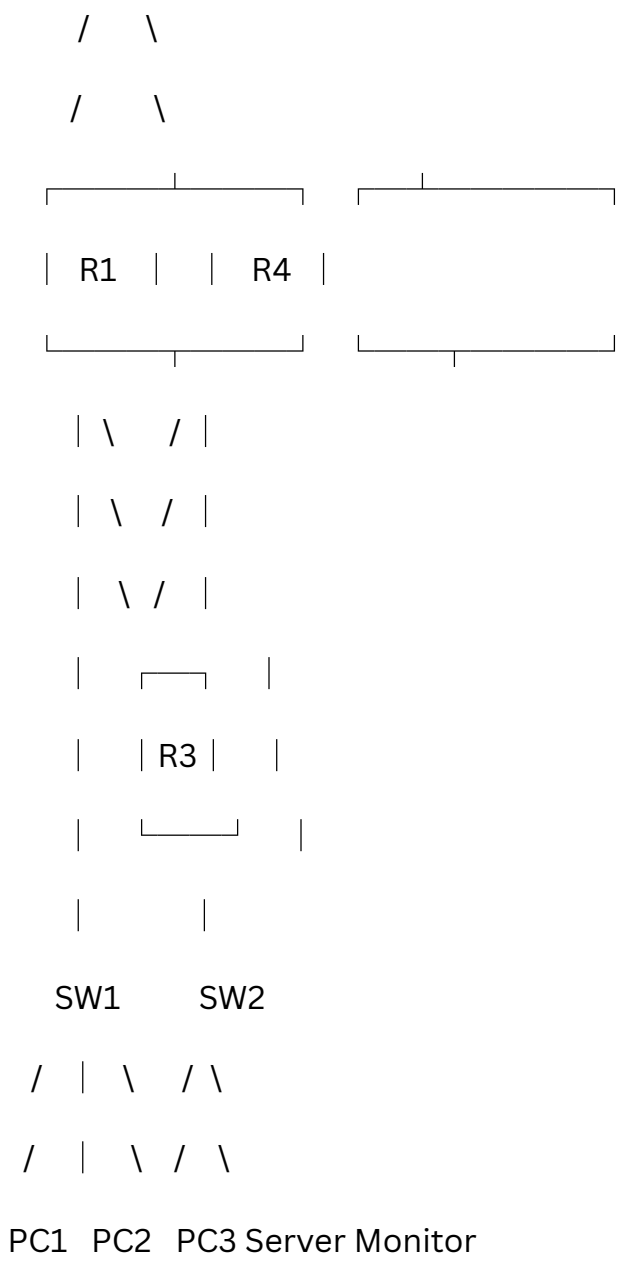




Client-side LAN:

Server-side LAN:

Complete logical topology:



4. Devices Used

Device	Quantity	Purpose
Cisco Router 2911	4	Routing and OSPF
Cisco Switch 2960	2	LAN connectivity
PC	3	DNS tunnel clients
Server	1	DNS/Tunnel server
PC	1	Network monitoring/testing

Router names:

R1

R2

R3

R4

Switch names:

SW1

SW2

Client names:

PC1

PC2

PC3

Server:

Server

Monitoring endpoint:

Monitor-PC

5. Port Connections

R1 Connections

R1 Interface	Connected Device	Device Interface	Purpose
G0/0	SW1	Fa0/1	Client LAN
G0/1	R2	G0/0	R1-R2 link
G0/2	R3	G0/0	R1-R3 link

R2 Connections

R2 Interface	Connected Device	Device Interface	Purpose
G0/0	R1	G0/1	R1-R2 link
G0/1	R4	G0/0	R2-R4 link

R3 Connections

R3 Interface	Connected Device	Device Interface	Purpose
G0/0	R1	G0/2	R1-R3 link
G0/1	R4	G0/1	R3-R4 link

R4 Connections

R4 Interface	Connected Device	Device Interface	Purpose
G0/0	R2	G0/1	R2-R4 link
G0/1	R3	G0/1	R3-R4 link
G0/2	SW2	Fa0/1	Server LAN

SW1 Connections

SW1 Interface	Connected Device	Device Interface
Fa0/1	R1	G0/0
Fa0/2	PC1	Fa0
Fa0/3	PC2	Fa0
Fa0/4	PC3	Fa0

SW2 Connections

SW2 Interface	Connected Device	Device Interface
Fa0/1	R4	G0/2
Fa0/2	Server	Fa0
Fa0/3	Monitor-PC	Fa0

6. IP Addressing Plan

Router-to-Router Networks

We use /30 networks for point-to-point router links.

R1 ↔ R2

Network: 10.0.12.0/30

Device	Interface	IP Address
R1	G0/1	10.0.12.1
R2	G0/0	10.0.12.2

R1 ↔ R3

Network: 10.0.13.0/30

Device	Interface	IP Address
R1	G0/2	10.0.13.1
R3	G0/0	10.0.13.2

R2 ↔ R4

Network: 10.0.24.0/30

Device	Interface	IP Address
R2	G0/1	10.0.24.1
R4	G0/0	10.0.24.2

R3 ↔ R4

Network: 10.0.34.0/30

Device	Interface	IP Address
R3	G0/1	10.0.34.1
R4	G0/1	10.0.34.2

7. Client LAN

Network:

192.168.10.0/24

Gateway:

192.168.10.1

Device	IP Address	Subnet Mask	Gateway
R1 G0/0	192.168.10.1	255.255.255.0	—
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC2	192.168.10.12	255.255.255.0	192.168.10.1
PC3	192.168.10.13	255.255.255.0	192.168.10.1

All three client PCs use **static IPv4 addresses**.

8. Server LAN

Network:

192.168.20.0/24

Gateway:

192.168.20.1

Device	IP Address	Subnet Mask	Gateway
R4 G0/2	192.168.20.1	255.255.255.0	—
Server	192.168.20.10	255.255.255.0	192.168.20.1
Monitor-PC	192.168.20.20	255.255.255.0	192.168.20.1

The Server and Monitor-PC also use **static IPv4 addresses**.

9. Router Configuration Completed

R1

The following configuration has been completed:

enable

configure terminal

hostname R1

interface gigabitEthernet 0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.12.1 255.255.255.252

no shutdown

exit

interface gigabitEthernet 0/2

ip address 10.0.13.1 255.255.255.252

no shutdown

exit

end

write memory

R2

enable

configure terminal

hostname R2

interface gigabitEthernet 0/0

ip address 10.0.12.2 255.255.255.252

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.24.1 255.255.255.252

no shutdown

exit

end

write memory

R3

enable

configure terminal

hostname R3

interface gigabitEthernet 0/0

ip address 10.0.13.2 255.255.255.252

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.34.1 255.255.255.252

no shutdown

exit

end

write memory

R4

enable

configure terminal

hostname R4

interface gigabitEthernet 0/0

ip address 10.0.24.2 255.255.255.252

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.34.2 255.255.255.252

no shutdown

exit

interface gigabitEthernet 0/2

ip address 192.168.20.1 255.255.255.0

no shutdown

exit

end

write memory

10. Configuration Verification

We used:

show ip interface brief

to verify the router interfaces.

The required interfaces are currently operational.

R1

G0/0 192.168.10.1 up/up

G0/1 10.0.12.1 up/up

G0/2 10.0.13.1 up/up

R2

G0/0 10.0.12.2 up/up

G0/1 10.0.24.1 up/up

R3

G0/0 10.0.13.2 up/up

G0/1 10.0.34.1 up/up

R4

G0/0 10.0.24.2 up/up

G0/1 10.0.34.2 up/up

G0/2 192.168.20.1 up/up

Unused interfaces such as R3 G0/2 and VLAN1 remain administratively down and are not part of the topology.

11. Connectivity Tests Completed

PC1 → R1

Command:

ping 192.168.10.1

Result:

Packets: Sent = 4

Received = 4

Lost = 0 (0% loss)

Therefore the PC1 → SW1 → R1 connection is working.

PC2 → R1

Command:

ping 192.168.10.1

Result:

Packets: Sent = 4

Received = 4

Lost = 0 (0% loss)

Therefore the PC2 → SW1 → R1 connection is working.

PC3 → R1

Configured with:

IP: 192.168.10.13

Mask: 255.255.255.0

Gateway: 192.168.10.1

LAN connectivity verified successfully.

Server → R4

Command:

ping 192.168.20.1

Result:

Packets: Sent = 4

Received = 4

Lost = 0 (0% loss)

Therefore the Server → SW2 → R4 connection is working.

12. Current Phase 1 Status

Completed

- Cisco topology created
- Four routers added
- Two switches added
- Three client PCs added
- Server added
- Monitor-PC added
- Router-to-router connections established
- LAN connections established
- IP addressing plan implemented
- Router interfaces enabled
- Client IP configuration completed
- Server IP configuration completed

- LAN connectivity verified

Pending

- Configure OSPF
- Verify OSPF neighbors
- Verify OSPF routing tables
- Test client-to-server connectivity through routers
- Determine normal OSPF path
- Simulate R2-R4 link failure
- Verify OSPF convergence
- Verify alternate R1 → R3 → R4 path
- Restore failed link
- Perform final Phase 1 connectivity tests

13. Next Step

The next stage is **OSPF configuration**.

We'll configure:

R1 → OSPF Area 0

R2 → OSPF Area 0

R3 → OSPF Area 0

R4 → OSPF Area 0

Then verify:

show ip ospf neighbor

show ip route

show ip ospf

Finally we'll deliberately shut down:

R2 G0/1

to simulate:

R2 ——— R4

X

and verify that the network switches to:

R1 → R3 → R4